

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-8: Fourier Transform

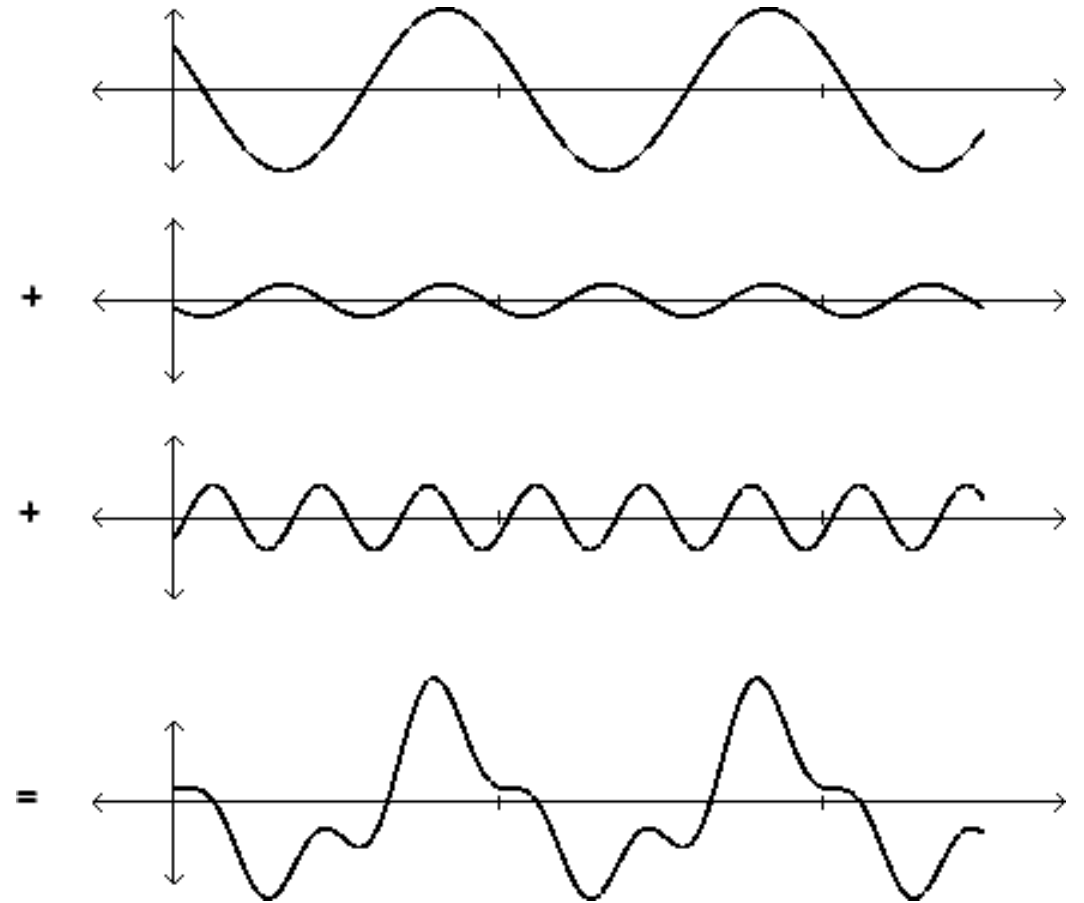
Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-4)

Md. Ariful Islam
Assistant Professor
Dept. of Robotics and Mechatronics Engineering
University of Dhaka

8. Fourier Series and Transform

Any periodic function can be expressed as the sum of sines and/or cosines of different frequencies, each multiplied by a different coefficient (we now call this sum a Fourier series).

It does not matter how complicated the function is; if it is periodic and satisfies some mild mathematical conditions, it can be represented by such a sum.



Even functions that are not periodic (but whose area under the curve is finite) can be expressed as the integral of sines and/or cosines multiplied by a weighing function.

The formulation in this case is the Fourier transform, and its utility is even greater than the Fourier series in many theoretical and applied disciplines.

We will be dealing only with functions (images) of finite duration, so the Fourier transform is the tool in which we are interested.

The material in the following section introduces the Fourier transform and the frequency domain.

8.1 Preliminary Concepts

8.1.1 Complex Numbers

A complex number, C , is defined as $C = R + jI$

where R and I are real numbers, and j is an imaginary number equal to the square of -1 that is

$$j = \sqrt{-1}$$

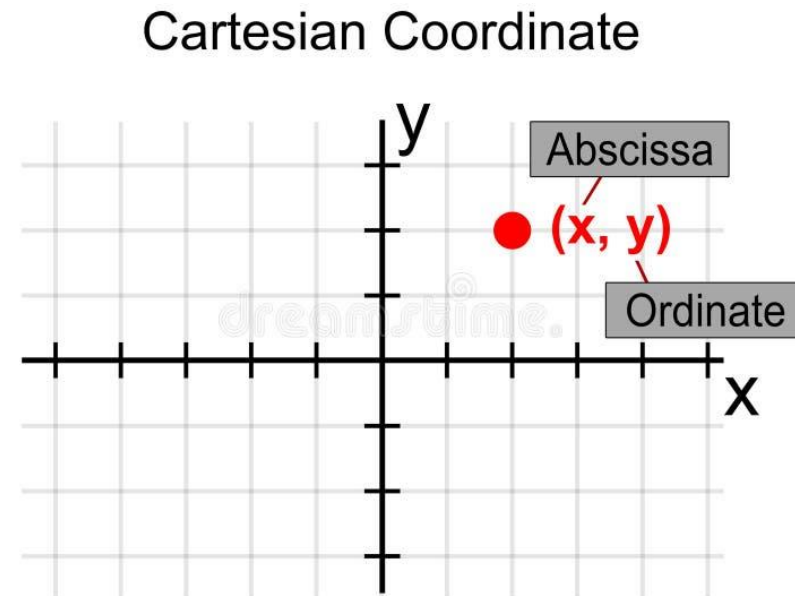
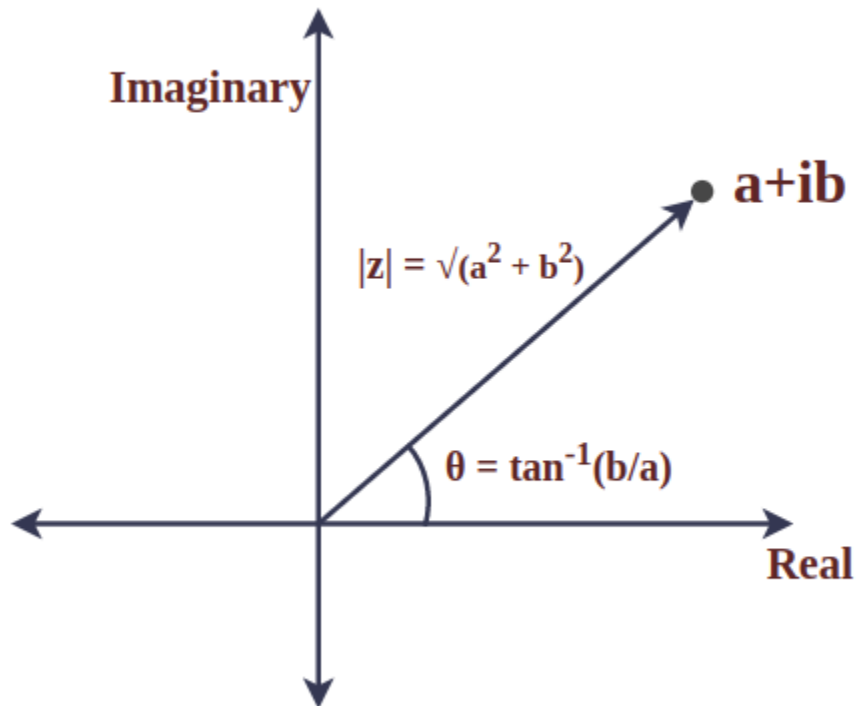
Here, R denotes the real part of the complex number and I its imaginary part.

Real numbers are a subset of complex numbers in which the conjugate of a complex number C , denoted C^* is defined as

$$C^* = R - jI$$

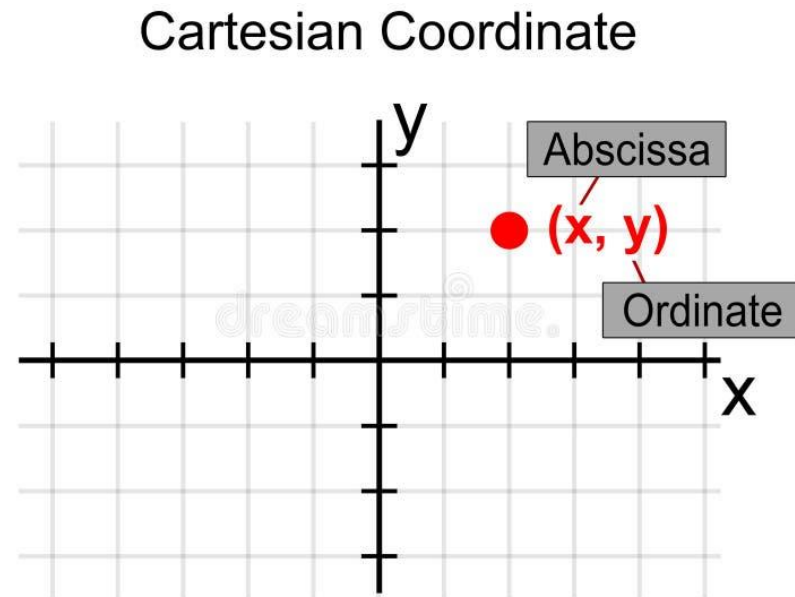
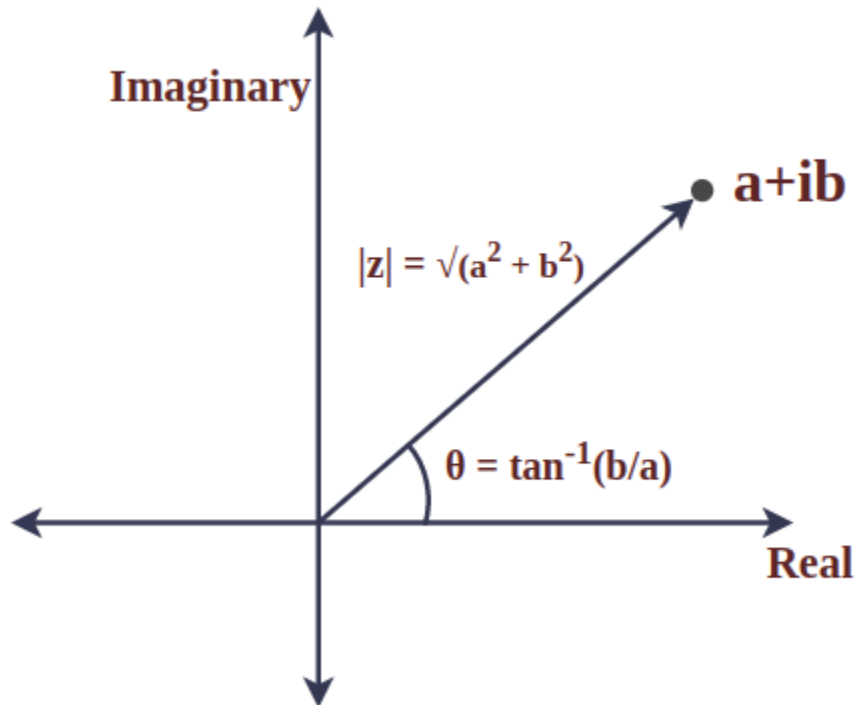
Complex numbers can be viewed geometrically as points in a plane (called the complex plane) whose abscissa is the real axis (values of R) and whose ordinate is the imaginary axis (values of I).

That is, the complex number is point (R, I) in the rectangular coordinate system of the complex plane.



The real axis represents things we measure directly (temperature, distance, money).

The imaginary axis represents abstract or phase-related concepts (AC signals, oscillations).



Sometimes, it is useful to represent complex numbers in polar coordinates

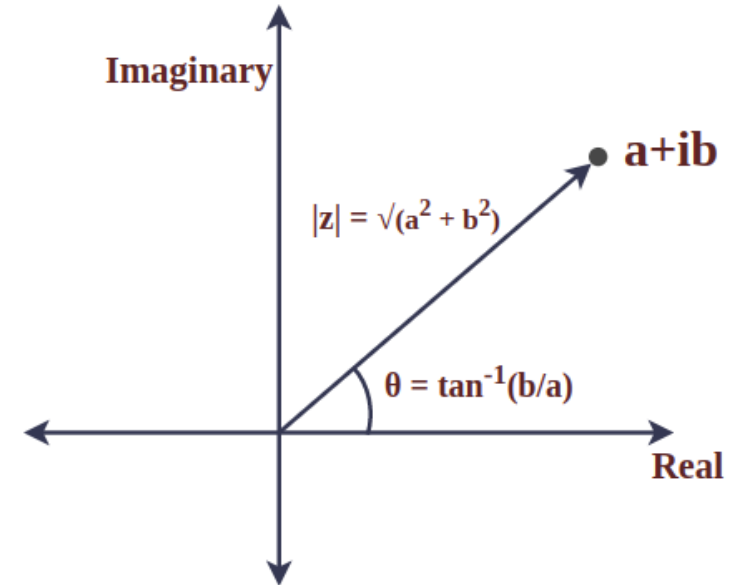
$$C = |C|(\cos \theta + j \sin \theta)$$

Where $|C| = \sqrt{R^2 + I^2}$ is the length of the vector extending from the origin of the complex plane to point (R, I), and θ is the angle between the vector and the real axis.

Drawing a simple diagram of the real and complex axes with the vector in the first quadrant will reveal that

$$\tan \theta = (I/R) \text{ or } \theta = \arctan(I/R)$$

The arctan function returns angles in the range $[-\pi/2, \pi/2]$



However, because *I and R can be positive and negative independently*, we need to be able to *obtain angles in the full range* $[-\pi, \pi]$

Using Euler's formula

$$e^{j\theta} = \cos \theta + j \sin \theta$$

Where $e = 2.71828\dots$, gives the following familiar representation of complex numbers in polar coordinates

$$C = |C|e^{j\theta}$$

Example: A complex function, $F(u)$, of a variable u , can be expressed as the sum

$$F(u) = R(u) + jI(u)$$

where $R(u)$ and $I(u)$ are the real and imaginary component functions.

the magnitude is

$$|F(u)| = \sqrt{R(u)^2 + I(u)^2}$$

the angle is

$$\theta(u) = \arctan[I(u)/R(u)]$$

8.1.2 Fourier Series

A function $f(t)$ of a continuous variable t that is periodic with period, T , can be expressed as the sum of sines and cosines multiplied by appropriate coefficients. This sum, known as a Fourier series, has the form

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{j\frac{2\pi n}{T}t}$$

Which is an expansion of sines and cosines follows from Euler's formula

$$e^{j\theta} = \cos \theta + j \sin \theta$$

where

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-j\frac{2\pi n}{T}t} dt \quad \text{for } n = 0, \pm 1, \pm 2, \dots$$

are the coefficients.

8.1.3 Impulses and Their Sifting Property

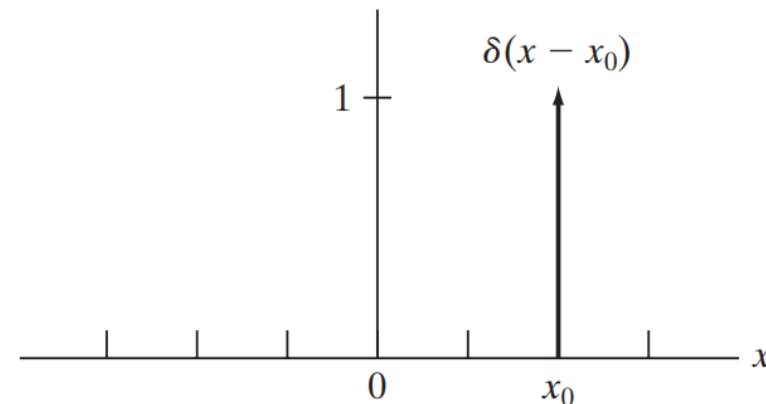
$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0 \end{cases}$$

and is constrained also to satisfy the identity $\int_{-\infty}^{\infty} \delta(t) dt = 1$

Physically, if we interpret t as time, an impulse may be viewed as a spike of infinity amplitude and zero duration, having unit area.

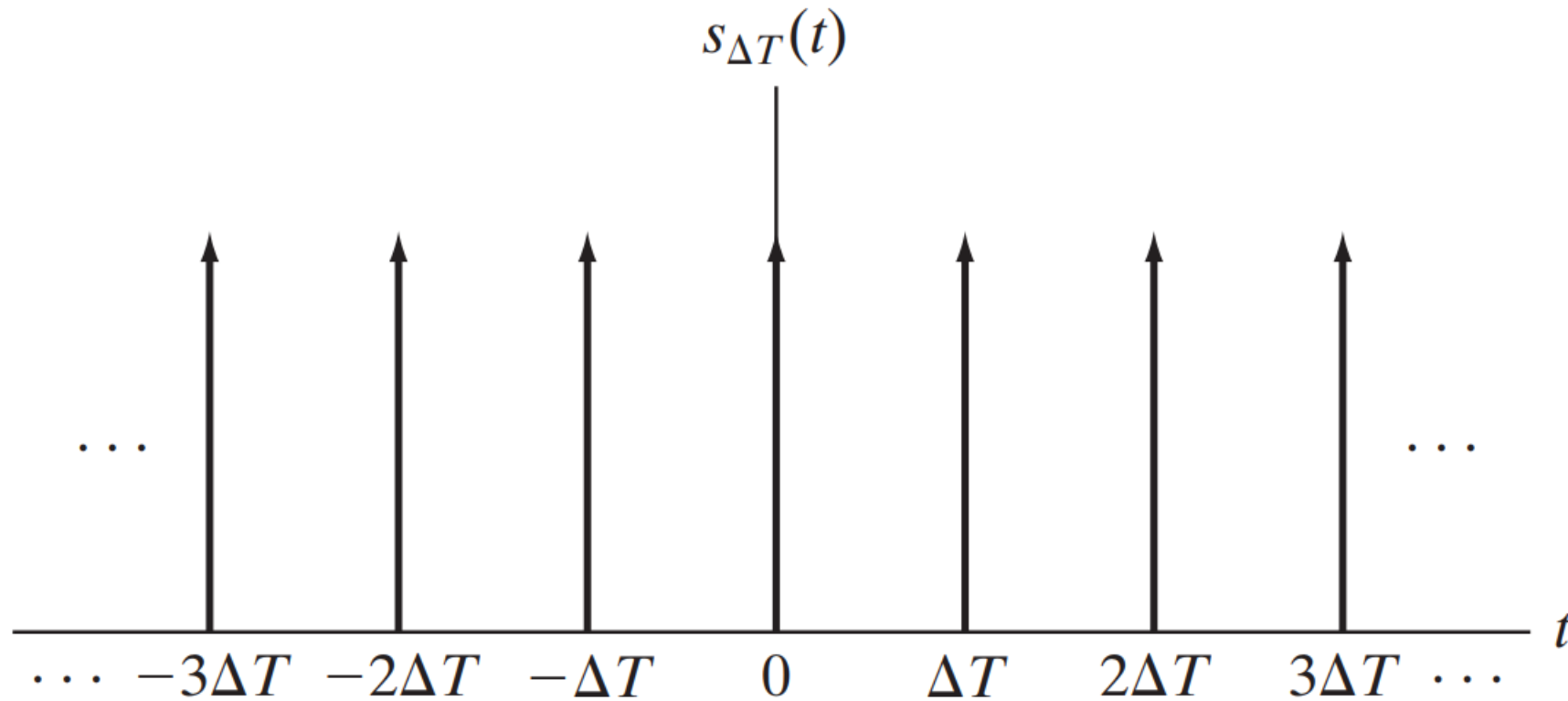
An impulse has the so called sifting property with respect to integration

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0)$$



Impulse train is defined as the sum of infinitely many periodic impulses ΔT units apart

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} \delta(t - n\Delta T)$$



8.2 The Fourier Transform of Functions of One Continuous Variable

$$\mathfrak{F}\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt$$

Where μ is also a continuous variable. Because t is integrated out, $\mathfrak{F}\{f(t)\}$ is a function only of μ

$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt$$

$$F(\mu) = \int_{-\infty}^{\infty} f(t) [\cos(2\pi\mu t) - j \sin(2\pi\mu t)] dt$$

we can obtain (t) back using the inverse Fourier transform $f(t) = \mathfrak{F}^{-1}\{F(\mu)\}$

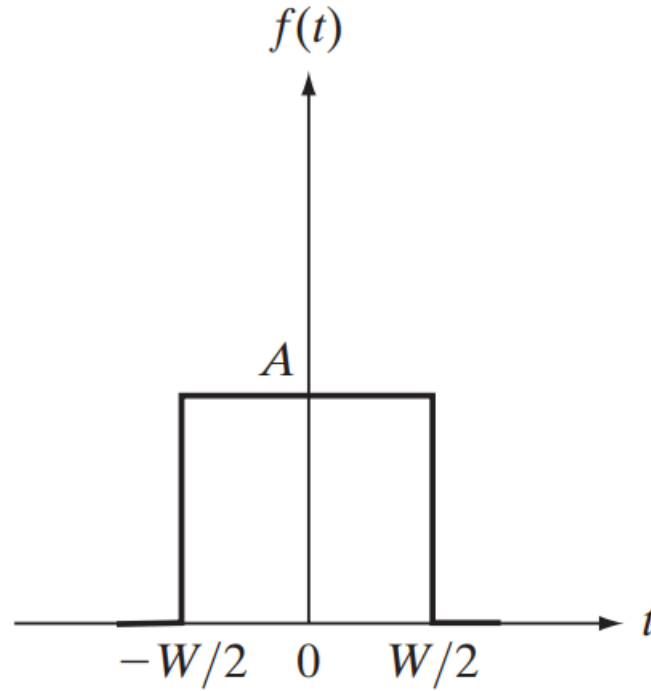
$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

t can represent any continuous variable, and the units of the frequency variable μ depend on the units of t.

For example, if t represents time in seconds, the units of μ are cycles/sec or Hertz (Hz). If t represents distance in meters, then the units of μ are cycles/meter, and so on.

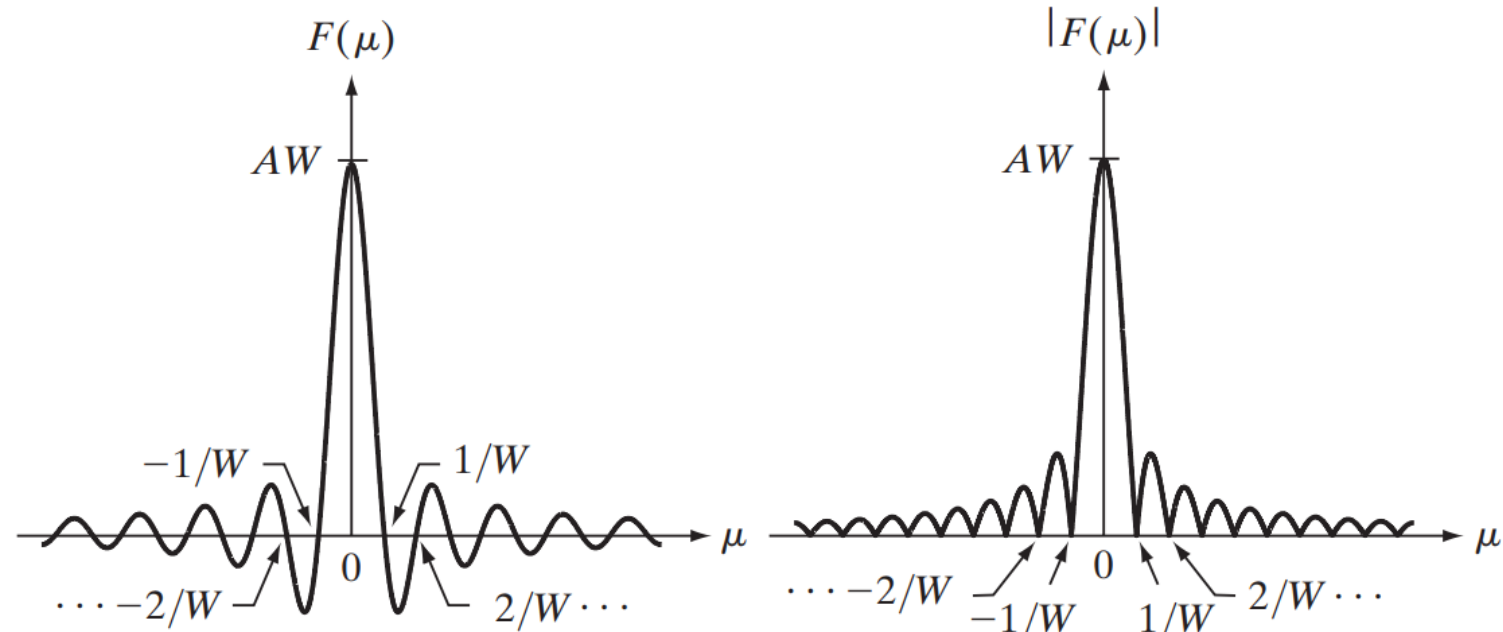
In other words, the units of the frequency domain are cycles per unit of the independent variable of the input function.

Example: Obtain the Fourier transform of the following function



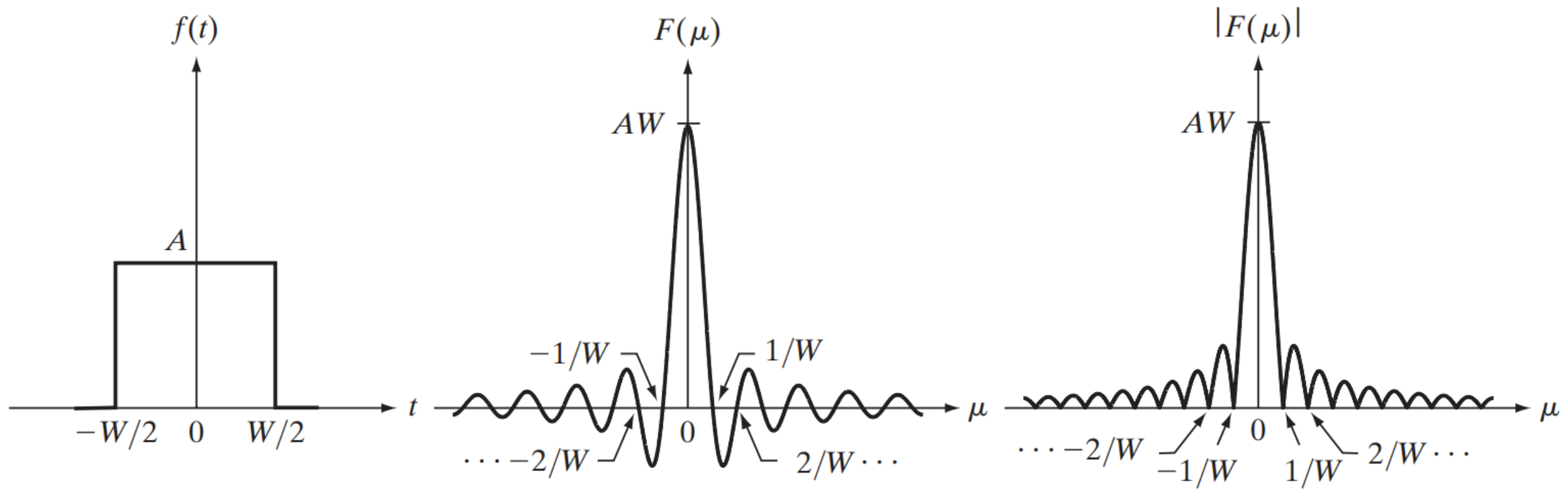
$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt$$

$$\begin{aligned}
 F(\mu) &= \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt = \int_{-W/2}^{W/2} A e^{-j2\pi\mu t} dt \\
 &= \frac{-A}{j2\pi\mu} \left[e^{-j2\pi\mu t} \right]_{-W/2}^{W/2} = \frac{-A}{j2\pi\mu} \left[e^{-j\pi\mu W} - e^{j\pi\mu W} \right] \\
 &= \frac{A}{j2\pi\mu} \left[e^{j\pi\mu W} - e^{-j\pi\mu W} \right] \\
 &= AW \frac{\sin(\pi\mu W)}{(\pi\mu W)}
 \end{aligned}$$



Fourier transform

Spectrum



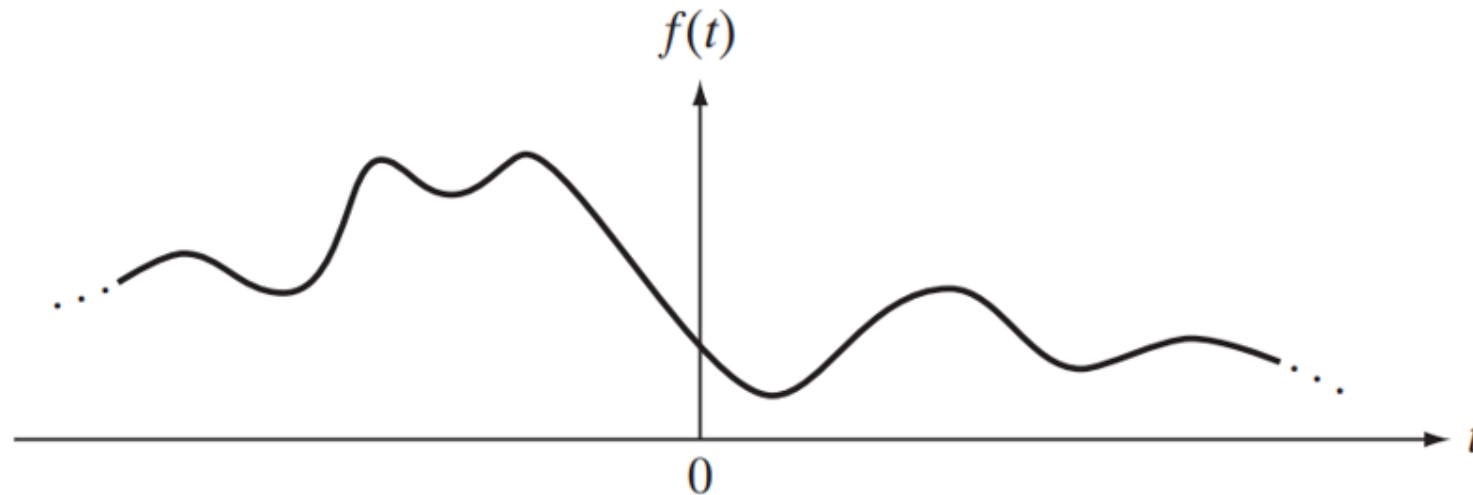
The locations of the zeros of both $F(\mu)$ and $|F(\mu)|$ are inversely proportional to the width, W , of the “box” function, that the height of the lobes decreases as a function of distance from the origin

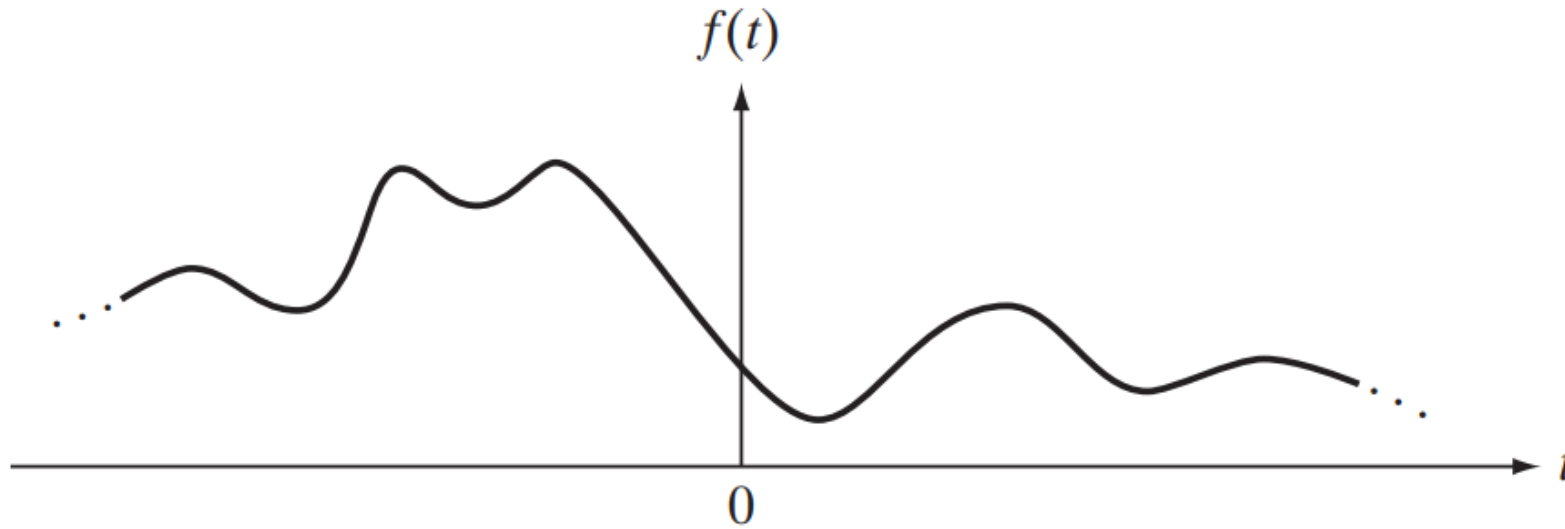
8.3 Sampling

Continuous functions have to be converted into a sequence of discrete values before they can be processed in a computer.

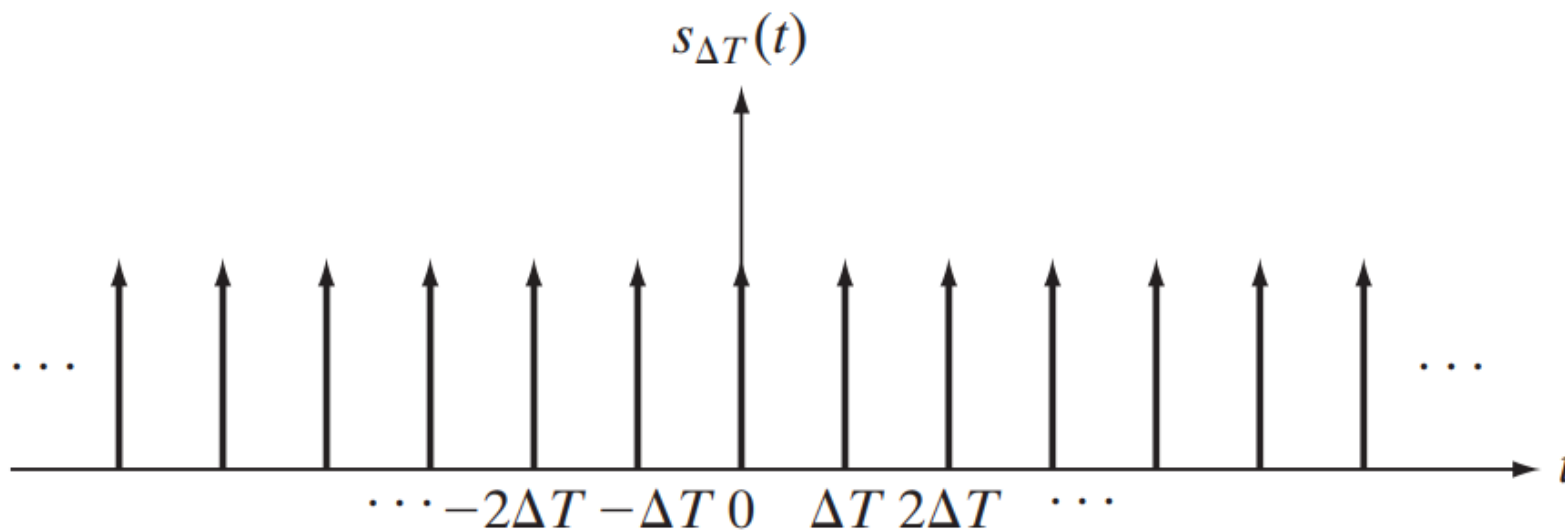
This is accomplished by using sampling and quantization.

Consider a continuous function, $f(t)$, that we wish to sample at uniform intervals (ΔT) of the independent variable t .





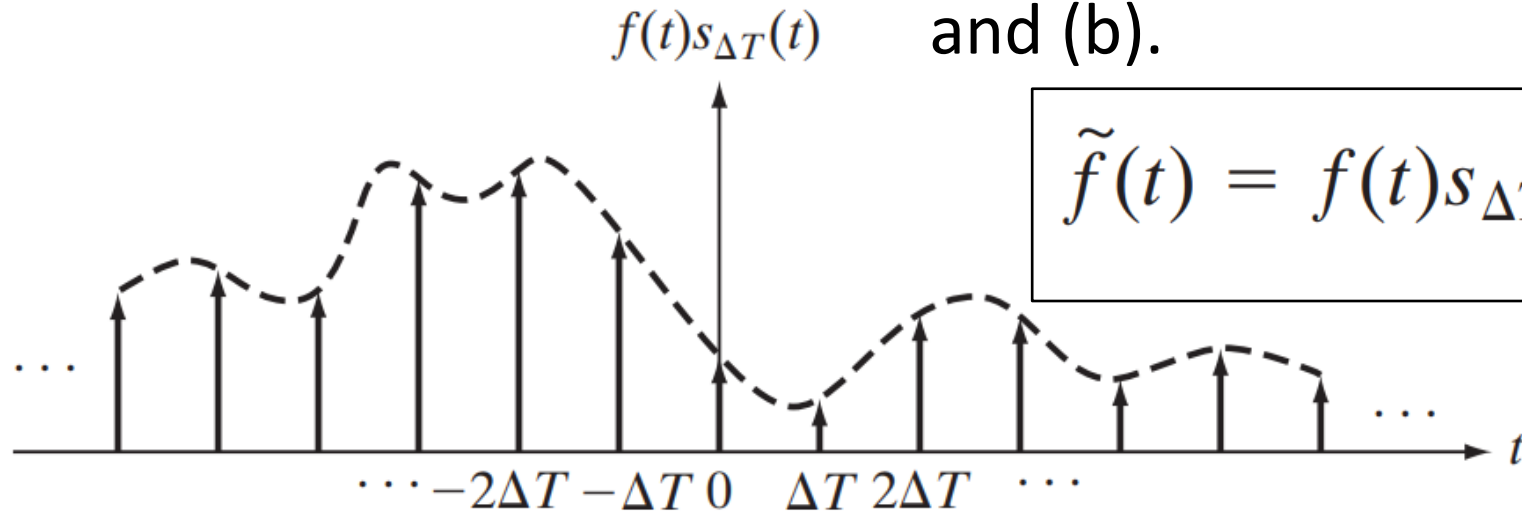
(a) A continuous function



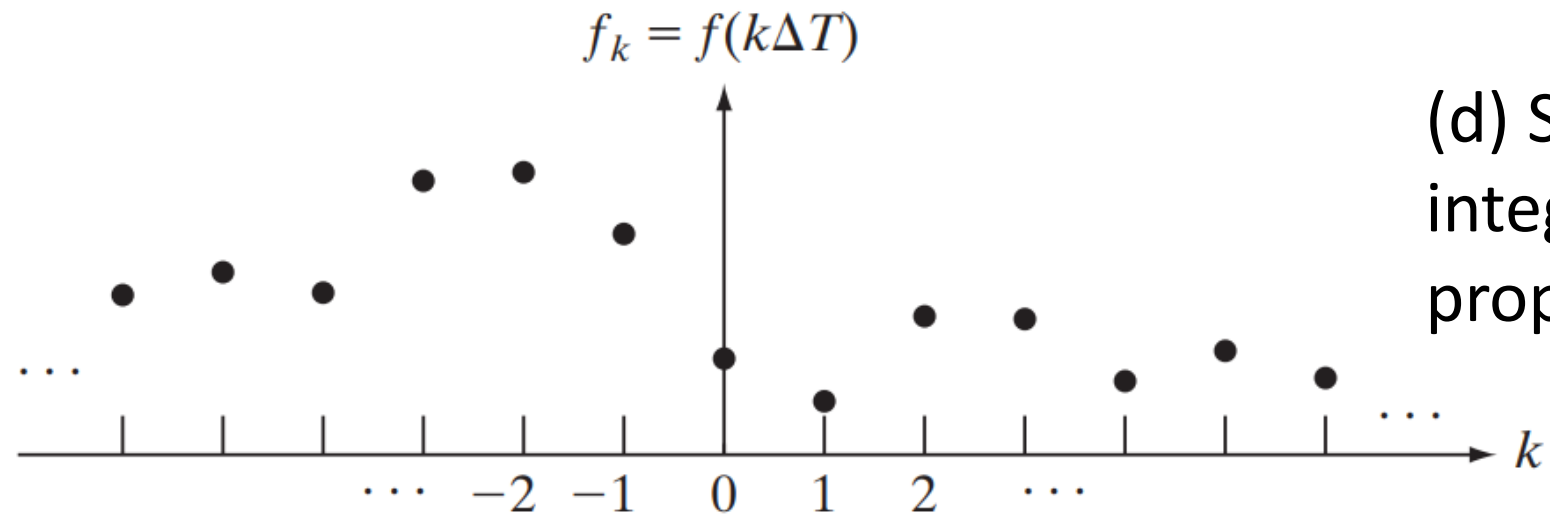
(b) Train of impulses used to model the sampling process.

$$s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} \delta(t - n\Delta T)$$

(c) Sampled function formed as the product of (a) and (b).



$$\tilde{f}(t) = f(t)s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\Delta T)$$



(d) Sample values obtained by integration and using the sifting property of the impulse.

$$\begin{aligned} f_k &= \int_{-\infty}^{\infty} f(t)\delta(t - k\Delta T) dt \\ &= f(k\Delta T) \end{aligned}$$

8.4 The Discrete Fourier Transform (DFT) of One Variable

$$\tilde{F}(\mu) = \int_{-\infty}^{\infty} \tilde{f}(t) e^{-j2\pi\mu t} dt \quad \boxed{\tilde{f}(t) = f(t)s_{\Delta T}(t) = \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\Delta T)}$$

$$= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} f(t)\delta(t - n\Delta T) e^{-j2\pi\mu t} dt$$

$$= \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} f(t)\delta(t - n\Delta T) e^{-j2\pi\mu t} dt$$

$$= \sum_{n=-\infty}^{\infty} f_n e^{-j2\pi\mu n\Delta T}$$

Although f_n is a discrete function, its Fourier $\tilde{F}(\mu)$ is continuous and infinitely periodic with period $1/\Delta T$

Therefore, all we need to characterize $\tilde{F}(\mu)$ is one period, and sampling one period is the basis for the DFT

Suppose that we want to obtain M equally spaced samples of $\tilde{F}(\mu)$ taken over the period $\mu = 0$ to $\mu = 1/\Delta T$

This is accomplished by taking the samples at the following frequencies:

$$\mu = \frac{m}{M\Delta T} \quad m = 0, 1, 2, \dots, M-1$$

$$F_m = \sum_{n=0}^{M-1} f_n e^{-j2\pi mn/M} \quad m = 0, 1, 2, \dots, M-1$$

This expression is the discrete Fourier transform we are seeking.

8.5 Inverse Discrete Fourier Transform (IDFT)

$$f_n = \frac{1}{M} \sum_{m=0}^{M-1} F_m e^{j2\pi mn/M} \quad n = 0, 1, 2, \dots, M-1$$

We used m and n in the preceding development to denote discrete variables because it is typical to do so for derivations.

However, it is more intuitive, especially in two dimensions, to use the notation x and y for image coordinate variables and u and v for frequency variables

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M} \quad u = 0, 1, 2, \dots, M-1$$

$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M} \quad x = 0, 1, 2, \dots, M-1$$

The Fourier Transform (FT) and its inverse should satisfy:

$$f(t) \xleftrightarrow{\text{Fourier Transform}} F(\omega)$$

where $f(t)$ and $F(\omega)$ are transform pairs.

In the Inverse Fourier Transform, the factor $1/M$ (or $1/N$ in the discrete case) appears due to normalization, ensuring that the forward and inverse transforms are consistent.

8.6 Relationship Between the Sampling and Frequency Intervals

If $f(x)$ consists of M samples of a function $f(t)$ taken ΔT units apart, the duration of the record comprising the set $\{f(x)\}$, $x = 0, 1, 2, \dots, M - 1$ is

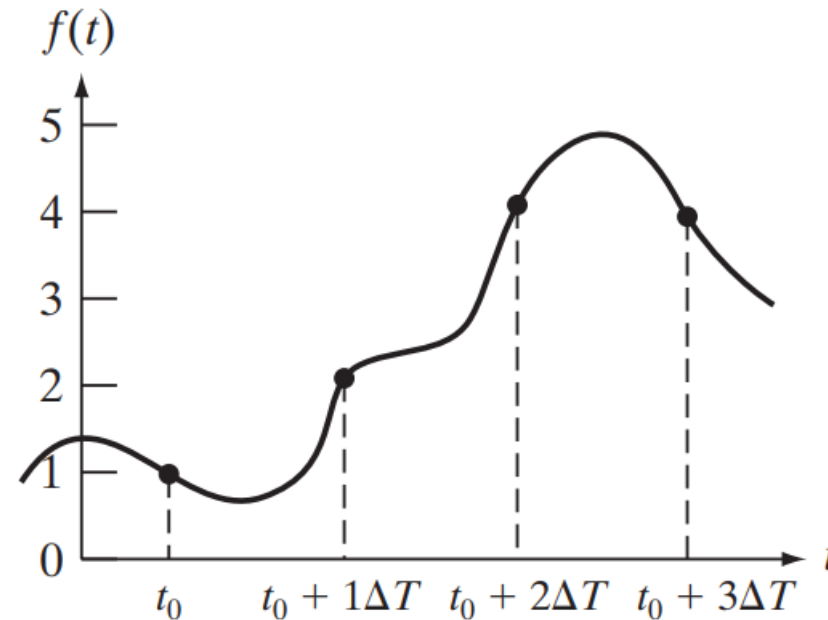
$$T = M\Delta T$$

The corresponding spacing Δu , in the discrete frequency domain follows

$$\Delta u = \frac{1}{M\Delta T} = \frac{1}{T}$$

8.7 The mechanics of computing the DFT

Example: Figure shows four samples of a continuous function, $f(t)$, taken ΔT units apart.



$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M} \quad u = 0, 1, 2, \dots, M-1$$

$$F(0) = \sum_{x=0}^3 f(x) = [f(0) + f(1) + f(2) + f(3)]$$

$$= 1 + 2 + 4 + 4 = 11$$

$$F(1) = \sum_{x=0}^3 f(x) e^{-j2\pi(1)x/4}$$

$$= 1e^0 + 2e^{-j\pi/2} + 4e^{-j\pi} + 4e^{-j3\pi/2} = -3 + 2j$$

$$F(2) = -(1 + 0j) \text{ and } F(3) = -(3 + 2j)$$

$$\begin{aligned}
 f(0) &= \frac{1}{4} \sum_{u=0}^3 F(u) e^{j2\pi u(0)} \\
 &= \frac{1}{4} \sum_{u=0}^3 F(u) \\
 &= \frac{1}{4} [11 - 3 + 2j - 1 - 3 - 2j] \\
 &= \frac{1}{4} [4] = 1
 \end{aligned}$$

8.8 2-D Discrete Fourier Transform and Its Inverse

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

Where $f(x, y)$ is a digital image of size $M \times N$

$u = 0, 1, 2, \dots, M - 1$ and $v = 0, 1, 2, \dots, N - 1$.

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

for $x = 0, 1, 2, \dots, M - 1$ and $y = 0, 1, 2, \dots, N - 1$

8.9 Relationships Between Spatial and Frequency Intervals

Suppose that a continuous function $f(t, z)$ is sampled to form a digital image, $f(x, y)$, consisting of $M \times N$ samples taken in the t- and z-directions, respectively.

Let ΔT and ΔZ note the separations between samples.

Then, the separations between the corresponding discrete, frequency domain variables are given by

$$\Delta u = \frac{1}{M \Delta T} \qquad \Delta v = \frac{1}{N \Delta Z}$$

Note that the separations between samples in the frequency domain are inversely proportional both to the spacing between spatial samples and the number of samples.

8.10 Fourier Spectrum and Phase Angle

Because the 2-D DFT is complex in general, it can be expressed in polar form:

$$F(u, v) = |F(u, v)|e^{j\phi(u,v)}$$

where the magnitude

$$|F(u, v)| = \left[R^2(u, v) + I^2(u, v) \right]^{1/2}$$

is called the Fourier (or frequency) spectrum

$$\phi(u, v) = \arctan \left[\frac{I(u, v)}{R(u, v)} \right]$$

is the phase angle.

Finally, the power spectrum is defined as

$$\begin{aligned} P(u, v) &= |F(u, v)|^2 \\ &= R^2(u, v) + I^2(u, v) \end{aligned}$$

R and I are the real and imaginary parts of $F(u, v)$ and all computations are carried out for the discrete variables

$$u = 0, 1, 2, \dots, M - 1 \qquad v = 0, 1, 2, \dots, N - 1$$

Therefore,

$|F(u, v)|$, $\phi(u, v)$, and $P(u, v)$ are arrays of size $M \times N$

Example: Given a 2x2 grayscale image with pixel values:

$$f(x, y) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Compute the 2D DFT $F(u, v)$.

$$F(u, v) = \sum_{x=0}^1 \sum_{y=0}^1 f(x, y) e^{-i2\pi \left(\frac{ux}{2} + \frac{vy}{2} \right)}$$

$$F(0, 0) = f(0, 0) + f(0, 1) + f(1, 0) + f(1, 1).$$

$$F(0, 1) = f(0, 0) + f(0, 1)e^{-i\pi} + f(1, 0) + f(1, 1)e^{-i\pi}$$

Repeat for $F(1, 0)$ and $F(1, 1)$

Magnitude Spectrum for a 3x3 Image

$$f(x, y) = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}$$

calculate the magnitude spectrum for $F(u, v)$ at all points.

$$F(u, v) = \sum_{x=0}^2 \sum_{y=0}^2 f(x, y) e^{-i2\pi \left(\frac{ux}{3} + \frac{vy}{3} \right)}$$

$$|F(u, v)| = \sqrt{\text{Re}(F(u, v))^2 + \text{Im}(F(u, v))^2}.$$

